

RESTful API for a Library Management System

Overview

The primary objective of the Library Management API is to provide a comprehensive and efficient backend solution for managing library operations. This includes handling user authentication and authorization, managing book and resource inventories, and facilitating the borrowing and returning of items. The API aims to streamline these processes, making it easier for both administrators and users to interact with the library system.

Scope:

The scope of this project includes the development of a RESTful API that supports the following functionalities:

- **User Authentication and Authorization:**
 - **Registration:** Allow users to create new accounts.
 - **Login:** Authenticate users and issue session tokens for secure access.
 - **Admin Privileges:** Restrict certain operations to administrators, such as creating, updating, and deleting books and resources.
- **Book Management:**
 - **Retrieve Books:** Fetch lists of books, search by various criteria (ID, title, genre), and get detailed information on individual books.
 - **Create, Update, Delete:** Allow administrators to add new books, update existing book details, and remove books from the inventory.
 - **Borrowing and Returning:** Enable users to borrow and return books, with real-time updates to availability.
- **Resource Management:**
 - **Retrieve Resources:** Fetch lists of resources, search by various criteria (ID, type, location), and get detailed information on individual resources.
 - **Create, Update, Delete:** Allow administrators to add new resources, update existing resource details, and remove resources from the inventory.
 - **Borrowing and Returning:** Enable users to borrow and return resources, with real-time updates to availability.
- **Deployment and Scalability:**
 - The API is designed to be deployed on cloud platforms like Vercel, ensuring scalability and reliability.

Team

Mani Setharasipalli - All CRUD operations
Anirudh Kalyani - Authentication and Authorization
Vivek Sunil Gaikwad - Error handling and validation
Ansh Srivastava - Testing
Maneesh Raja - Data Modeling
Kakkad Manav Bharatbhai - Deployment
Gulzar Hussain Jaffari - Testing and Documentation

Technologies

Node.js: Server-side JavaScript runtime.
Express.js: Web framework for building RESTful APIs.
MongoDB: NoSQL database for data storage.
Mongoose: ODM library for MongoDB.
Crypto : Session-based authentication.
express-validator: Middleware for validating request data.
dotenv: Module for environment variable management.
Vercel: Platform for deployment.
Jest and Supertest: Api testing.
Postman: API testing and Documentation

Setup

Clone the repository.

```
git clone https://github.com/viv59/api-creation.git
```

Install dependencies using npm install.

```
npm install
```

Configure environment variables in a .env file (refer .env.example).

Running the Server:

```
npm run dev
```

Reference/Resources

Postman Documentation: <https://documenter.getpostman.com/view/31981526/2sA3kYhKCU>

Website: <https://libraryrestapi.vercel.app>

Github Repo: [Private Repo](#)

Risks

Deployment: sometime vercel may display serverless error to troubleshoot just reload the website

Database connectivity: Sometimes MongoDB may not access your location properly.

Thank You!